

Paddy Yield Prediction Using Ensemble and Traditional Regression Models With Feature Selection and SHAP Analysis

Team 5

1st Soham Datta

Bennett University

Greater Noida, India

e23cseu0585@bennett.edu.in

2nd Ahzam Khan

Bennett University

Greater Noida, India

e23cseu0613@bennett.edu.in

3rd Nitin Kumar Jha

Bennett University

Greater Noida, India

e23cseu0628@bennett.edu.in

4th Naman Garg

Bennett University

Greater Noida, India

e23cseu0584@bennett.edu.in

Abstract—Rice is one of the most important crops in India and millions of farmers depend on it for their livelihood. However, predicting how much paddy a farm will produce is difficult because yield depends on many factors like rainfall, temperature, soil type, fertilizer use and farming practices. In this study, we use machine learning models to predict paddy rice yield using a dataset of 2,790 samples from Tamil Nadu, India, covering 45 features related to climate, soil and agronomic inputs. We train and compare six regression models including Multiple Linear Regression, Ridge Regression, Support Vector Regression, Random Forest, XGBoost and LightGBM on two target variables: total Paddy Yield in Kg and Paddy Yield per Hectare. Feature selection was performed using Pearson Correlation Coefficient and Random Forest feature importance to reduce dimensionality from 45 to 10 and 24 features for the two targets respectively. For total yield prediction, all models performed well with Random Forest achieving the best results with a test R squared of 0.9911 and RMSE of 850.46. For yield per hectare, performance dropped across all models, with XGBoost and LightGBM achieving the highest test R squared of 0.5119. An interesting finding is that total yield prediction relied entirely on agronomic features while yield per hectare additionally depended on weather and categorical features, showing that normalizing the target changes which factors matter most. The study demonstrates that machine learning can provide accurate yield estimates that could help farmers, buyers and government agencies make better decisions about crop planning and food security.

Index Terms—

I. INTRODUCTION

In India agriculture is one of the major sources of livelihood for millions of people, and paddy (rice) is one of the most widely grown crops. Large number of farmers depend on paddy production for their livelihood. However the crop production is highly dependent on environmental factors such as rainfall, temperature, humidity, wind speed and soil type. However, in recent years, the imbalanced monsoon and climate change have made it difficult to forecast the level of production. Farmers here usually forecast their estimated production based on their vast past experiences, which may not be accurate. Thus Inaccurate forecasting may lead to inefficient storage planning, financial losses and market price fluctuations. So

there lies a need for a more accurate and systematic approach to forecast the crop production. The objective of this project is to predict paddy crop production using machine learning models or algos. By using a data-driven approach, we attempt to understand how different variables affect crop production and explore interesting patterns in the data set. By following a series of activities such as data preprocessing, feature selection, model development and evaluation we plan to develop an accurate prediction model. The primary goal is the estimation of paddy crop production based on environmental and crop-related factor and variables that are present in the dataset. Instead of Simply relying on traditional knowledge or manual estimates we would like to use data-driven approaches in understanding how various variables interact with each other to impact the crop production. By training various machine learning algorithm on the dataset and understanding which features or factors lead to the maximum production in growth we will be able to recognize patterns that are not immediately apparent through our simple analysis. Essentially by understanding how much paddy we can grow based on current farming practices and climate factors that helps not only the farmers but also the people who buy food and our government. If farmers understand how much production they will grow ahead of time they can make informed choices about water, fertilizers and best crop maintenance practices. If accurate estimates of the crop production are available in advance then the government can make better decisions regarding procurement, storage management, distribution planning etc. beforehand. In a country where food security farmer income are major topics ML in agriculture is not merely a academic pursuit but a solution to realworld problem.

II. RELATED WORK

Machine learning application to agricultural yield prediction has changed significantly over the past years, but the journey reveals as much about continued challenges as it does about tech progress.

Van Klompenburg et al. [1] analyzed 567 and found that temperature, rainfall, and soil type dominated as predictive

features, while Artificial Neural Networks and Convolutional Neural Networks emerged as the preferred architectures. Real issue however was identifying insufficient data, especially time-series data across different climates and crop types. This data scarcity would continue through nearly every subsequent study.

The same team [2] looked at 102 papers and found that while individual architectures like CNN and LSTM performed well, hybrid networks that combined different approaches (CNN-LSTM, CNN-RNN) consistently outperformed because such networks suited the temporal patterns in crop yield data with their temporal-based architecture and feedback loops. Despite these, models struggled to generalize across different crops and environments.

Liu et al. [3] used multiple data sources, combining SAR and optical remote sensing with biophysical parameters. Their findings across two crop seasons revealed that context matters a lot. Summer rice responded best to XGBoost across all growth stages, but kharif rice needed different models at different times (XGBoost at 45 days, Neural Networks at 60 days, Cubist at 90 days). This approach worked well but showed that the sophisticated remote sensing infrastructure required put it out of reach for most farmers, especially in developing regions where precision agriculture matter most.

Muthukumaran et al. [4] built a hybrid model that used five different feature selection methods, and merged results to identify crucial features for paddy yield. More importantly, they moved beyond prediction to include variety recommendations based on farmer preferences.

Zhang et al. [5] achieved accuracy of 95.33% using hyperspectral imaging and CNN-Transformer models, identifying spectral bands that distinguished rice varieties. However, hyperspectral imaging equipment costs more than most farms' annual revenues. Wang et al. [6] brought costs down by using UAV-based multispectral sensing, achieving AUC of 0.804 with more accessible technology. Yet even drone-based approaches required specialized equipment and expertise, and performance varied significantly across growth stages, raising questions about real-world reliability.

A recent comprehensive study [7] trained models on 46,500 images across 62 rice varieties and actually releasing them through a web platform and mobile app for real-time variety identification. This represented shift from laboratory to field. But farmers don't just need to identify varieties; they need to know which variety to plant, how much fertilizer to use, what yields to expect.

Recognizing this, Patel & Patel [8] developed AgriRec, which combined crop and fertilizer recommendations in a single system. Testing on 5,000 land samples from Gujarat covering 24 crops, they achieved strong accuracies (95.85% for crops, 92.11% for fertilizers). However the reliance on traditional ML algorithms raised doubts about handling the complex interactions and class imbalances that are common in agricultural data.

TabNet [9], a DL architecture designed for tabular data provided built-in explainability for farmers who need to un-

derstand why a recommendation was made, not just what it is. IoT sensors integrated allowed for real-time soil and climate monitoring. The approach handled complex feature interactions that stymied traditional ML, but the computational requirements remained high for smallholder farmers.

Govindwar et al. [10] had high accuracies (99.31% for crops, 90% for fertilizers) with ensemble methods like XGBoost and Random Forest. Their system showed that near-perfect performance was possible using readily available features: soil nutrients, pH, temperature, humidity, rainfall. The practical benefits are: intelligent recommendations saved farmers time and money compared to traditional trial and error. Yet this work treated crop and fertilizer recommendations as separate problems rather than the unified, interdependent optimization they actually are. And like most existing systems, it lacked variety-specific recommendations.

The path forward is simple from general to crop-specific methods from single-source to multi-modal learning, from prediction to prescriptive systems, from black-box to interpretable AI. There is still no integrated system capable of yield prediction and variety and input recommendation. High accuracy still requires infrastructure beyond the reach of most farmers. And interpretability, a key to adoption, is difficult to achieve without compromising accuracy. The integration of advanced feature selection techniques, ensemble learning, and explainable AI provides promise, especially for developing environments where accurate intelligence could make a direct difference in food security.

III. METHODOLOGY

The objective of this study is testing the prediction of paddy yield using various traditional and advanced machine-learning model, and also analyzing and identifying which agronomic or climate features are influential to paddy yield using feature selection statistical methods and SHAP explainability (SHapley Additive exPlanations). Importantly, two target formulations were chosen for model training and feature analysis: Paddy yield(in Kg) and Paddy yield per Hectare.

A. Feature Selection

Given the dataset's massive 45 predictor features, ranging from farm inputs, agronomic treatments, and weather/climate conditions, a feature selection pipeline was necessary to narrow down and reduce dimensionality and preserve the most important features before model training.

1) *Pearson Correlation Coefficient*: Pearson Correlation Coefficient was used as an initial filter to understand the relationship between each numerical feature and the target variables. It can be represented by the formula below:

$$r = \frac{n(\sum xy) - (\sum x)(\sum y)}{\sqrt{[n \sum x^2 - (\sum x)^2][n \sum y^2 - (\sum y)^2]}} \quad (1)$$

Equation (1) calculates the covariance of variables and normalizes it by their standard deviations. The variable n represents number of data points, x represents the numerical features, while y represents the target variable.

For this study, features with a correlation magnitude below a pre-defined threshold (in this case a threshold of 0.01) were chosen for removal to reduce noise and preserve linearly predictive features.

2) *Random Forest Feature Importance*: Random Forest Regressor model provides a built-in feature importance mechanism that ranks features based on Mean Decrease in Impurity (MDI) scores that measure contribution to variance reduction across all decision trees in the ensemble. In this study, the Pearson-filtered numerical features were combined with the one-hot encoded categorical features (not used in Pearson Correlation Coefficient) and passed on to the Random Forest Regressor. Features that had very little to zero importance scores were removed from the predictor variables to form a finalized predictor variable subset ready for further regression model training. Similar to the previous feature selection method, a RF feature importance threshold was chosen to be 0.01. Any features with values below this score threshold were omitted.

B. Machine Learning Models

A total of six regression models were trained and evaluated onto the dataset. Three of the models are traditional statistical machine learning models, while the other three are more advanced, ensemble based models. This comparison allows for assessing whether the complexity of tree-based ensembles offers meaningful gains on the agriculture-based tabular dataset. The model description used in this study are provided in this section below.

1) *Multiple Regression*: Linear regression algorithm finds a relationship between the independent and dependent variables by fitting a line that minimizes the differences between observed data points and the predicted values on the line. Multiple regression is a type of linear regression that allows more than one independent variable to be included, which was used in this study. The formula is as follows:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n + \epsilon \quad (2)$$

In Equation (2) y is the predicted paddy yield, $x_1, x_2, \dots, x_n$ represent the n distinct independent features, β_0 is the y-intercept, $\beta_1, \beta_2, \dots, \beta_n$ are the regression coefficients for each feature, and ϵ is the residual error.

2) *Ridge Regression*: Ridge Regression algorithm can be considered an extension of the linear regression model but adds a penalty term that is proportional to the squared magnitudes of the coefficients. This shrinks large coefficients and takes care of multicollinearity to provide more reliable predictions. The widely accepted formula is as follows:

$$\min_{\beta} (\|y - X\beta\|^2 + \lambda \|\beta\|^2) \quad (3)$$

In Equation (3), y is the vector of target values, X is the feature matrix, β is the vector of regression coefficients, and λ is the regularization parameter controlling the strength of the penalty term. $\|y - X\beta\|^2$ represents the squared residual errors, while $\|\beta\|^2$ represents the squared L2 norm of the coefficients.

3) *Support Vector Regression*: Support Vector Regression algorithm works by fitting kernel functions to handle both non-linear and linear relationships in data. In this study, both linear and RBF kernels were implemented for the model training depending on the two target variables. For predicting the raw total Paddy Yield target, a linear kernel was used:

$$f(x) = w^T x + b \quad (4)$$

For predicting the Paddy Yield per Hectare target, the Radial Basis Function (RBF) kernel was used:

$$f(x) = \sum_{i=1}^n (\alpha_i - \alpha_i^*) K(x_i, x) + b \quad (5)$$

where the RBF kernel $K(x_i, x)$ is mathematically defined as:

$$K(x_i, x) = \exp(-\gamma \|x_i - x\|^2) \quad (6)$$

Regardless of the chosen kernel, SVR model learns the best parameters by minimizing the following optimization objective:

$$\min \frac{1}{2} \|w\|^2 + C \sum_{i=1}^n (\xi_i + \xi_i^*) \quad (7)$$

The selection of different kernels for each target is due to the underlying distributions of data after pre-processing techniques. The linear kernel was for the normally distributed raw yield, while the RBF kernel better captured the non-linear structure which was introduced by the per-hectare normalization.

4) *Random Forest Regression*: Random Forest Regression is an advanced ensemble learning model that works by combining multiple decision trees to produce more accurate predictions. The predictions are obtained by averaging the predictions of the decision trees, also known as the Bootstrap Aggregation (bagging) technique.

$$\hat{y} = \frac{1}{B} \sum_{b=1}^B f_b(x) \quad (8)$$

where B is the total decision trees in the ensemble, and $f_b(x)$ is the continuous yield prediction made by the b -th individual decision regression tree.

5) *Extreme Gradient Boosting (XG Boost)*: XGBoost Algorithm is an advanced Gradient Boosting algorithm, which builds decision tree sequentially with each tree attempting to correct the mistakes made by the previous one. A base learner model predicts average of the target variable, after which the errors between predicted and actual values are calculated, after which the next tree is trained on the errors and attempts to correct the errors. This continues until a stopping criteria is met. It is an iterative process where the final prediction is the combined sum of predictions from all the trees. The final prediction can be represented as:

$$\hat{y}_i = \sum_{k=1}^K f_k(x_i) \quad (9)$$

where $\hat{y}_i$ is the final predicted value for the i -th data point, K is the total number of trees in the ensemble, and $f_k(x_i)$ represents the prediction of the k -th tree for the i -th data point.

6) *Light Gradient Boosting Machine (LightGBM)*: LightGBM, a framework developed by Microsoft, employs features such as Gradient-based One-Side Sampling (GOSS) and Exclusive Feature Bundling (EFB). It was implemented to test the leaf-wise growth strategy that may capture finer patterns in this dataset of 2,790 samples compared to level-wise approach of traditional gradient boosting,

C. Cross Validation and Hyperparameter Tuning

Importantly, K-fold Cross Validation was applied during model training to reduce risk of overfitting. Dataset was partitioned into five folds, in each iteration four folds were for training and the remaining was used for validation. Hyperparameter tuning was done through GridSearchCV. The optimal hyperparameter configurations identified for each model under both target variables are reported in Tables I and II.

TABLE I
OPTIMAL HYPERPARAMETERS FOR TARGET 1: PADDY YIELD (IN KG)

Model	Hyperparameter	Selected Value
Multiple Linear Regression	—	—
Ridge Regression	α (lambda)	0.01
SVR	Kernel	Linear
	C	10
	ϵ	0.01
Random Forest	n_estimators	300
	max_depth	10
	max_features	sqrt
	min_samples_leaf	1
	min_samples_split	2

TABLE II
OPTIMAL HYPERPARAMETERS FOR TARGET 2: PADDY YIELD PER HECTARE

Model	Hyperparameter	Selected Value
Multiple Linear Regression	—	—
Ridge Regression	α (lambda)	100
SVR	Kernel	RBF
	C	100
	ϵ	0.5
Random Forest	n_estimators	200
	max_depth	10
	min_samples_leaf	4
	min_samples_split	10
XGBoost	n_estimators	100
	max_depth	3
	learning_rate	0.05
LightGBM	n_estimators	100
	max_depth	3
	learning_rate	0.05
	num_leaves	15

IV. EXPERIMENTAL RESULTS

A. Dataset Description

The Paddy Dataset [11] (UCI Repository ID: 1186), dated July 14, 2025, submitted by Muthukumaran Subramanian et al., is taken from the UCI Machine Learning Repository

with a Creative Commons Attribution 4.0 International (CC BY 4.0) license. This dataset was chosen for its relevance to precision agriculture, reflecting actual paddy rice production data from Tamil Nadu, India, with 2,790 samples and 46 features (45 predictors and target), with no missing data, varied feature types (categorical, integer, continuous), and immediate relevance to yield prediction models in the face of global food security crises. It systematically captures farmer behavior such as land area (Hectares), variety (CO43, Ponmani), soil types (alluvial, clay), nursery types (dry/wet), seed rates, fertilizers (DAP at 20 days, Urea at 40 days), pesticides, and climatic factors for four growth stages (rainfall, temperature, wind, humidity), allowing complex ML experimentation, important feature exploration and identification, and explainability.

B. Chosen Features for Both Targets

Initially, the dataset was pre-processed by standard scaling continuous numerical features and one-hot encoding categorical features. One-hot encoded features were excluded from pearson correlation coefficient test and added back after the test retained top correlated features. Table III below shows the selected features after feature selection methods for both targets. An observation is that Target 1 retained only agronomic/farm features while Target 2 additionally captured weather and categorical features. This shows that the per-hectare normalization in Target 2 have reduced scale-dependent agronomic inputs, allowing features with possible non-linear relationships to surface.

C. Machine Learning Model Results

The Tables IV and V represent the results achieved by the various traditional and advanced machine learning models on the Tamil Nadu Paddy Dataset.

All models trained on Target variable 1 which is total ‘Paddy Yield (in Kgs)’ demonstrated exceptionally strong performance as seen in Table IV. Multiple linear regression and ridge regression produced near identical results (CV RMSE of 950.56 and test R^2 of 0.9889), showing that L2 penalty in ridge regression offered no meaningful correction over the linear model baseline. This could be because multicollinearity was already mitigated during feature selection. SVR model with linear kernel performed similar by getting a test R^2 of 0.9886 but a higher RMSE score of 963.63. Moving on to ensemble based learning, random forest regressor attained the lowest RMSE of all, with a score of 850.46, but the highest test R^2 of 0.9911. This confirms that the ensemble’s capacity to model feature interactions yielded a small but consistent gain over traditional linear methods. The almost negligible gap between random forest’s train and test score indicate the model generalized well without overfitting. Since these models already performed high, no other models were implemented.

Interestingly, performance of all models trained on Target 2, which is Paddy Yield per Hectare, declined, as can be visualized in Table V. This reflects the increased difficulty of predicting a normalized yield measure from the available features. Multiple Linear Regression and Ridge again produced

TABLE III
FEATURES SELECTED AFTER PEARSON CORRELATION AND RANDOM FOREST IMPORTANCE FILTERING

Target	Selected Feature	Category
Target 1 Paddy Yield (in Kg) $n = 10$	Micronutrients_70Days	Agronomic
	LP_Mainfield (in Tonnes)	Agronomic
	DAP_20days	Agronomic
	LP_nurseryarea (in Tonnes)	Agronomic
	Seedrate (in Kg)	Agronomic
	Nursery area (Cents)	Farm
	Pest_60Day (in ml)	Agronomic
	Weed28D_thiobencarb	Agronomic
	Urea_40Days	Agronomic
	Potassh_50Days	Agronomic
Target 2 Yield per Hectare $n = 24$	DAP_20days	Agronomic
	Pest_60Day (in ml)	Agronomic
	LP_Mainfield (in Tonnes)	Agronomic
	Urea_40Days	Agronomic
	Micronutrients_70Days	Agronomic
	Potassh_50Days	Agronomic
	Nursery area (Cents)	Farm
	Weed28D_thiobencarb	Agronomic
	Seedrate (in Kg)	Agronomic
	Variety_delux ponni	Categorical (OHE)
	Soil Types_clay	Categorical (OHE)
	Nursery_wet	Categorical (OHE)
	Variety_ponmani	Categorical (OHE)
	Max temp_D61_D90	Weather
	Relative Humidity_D1_D30	Weather
	Inst Wind Speed_D1_D30 (in Knots)	Weather
	Min temp_D31_D60	Weather
	Max temp_D1_D30	Weather
	Min temp_D61_D90	Weather
	Min temp_D1_D30	Weather
	Relative Humidity_D61_D90	Weather
	Inst Wind Speed_D61_D90 (in Knots)	Weather
	51_70DRain (in mm)	Weather
	Max temp_D31_D60	Weather

TABLE IV
MODEL PERFORMANCE — TARGET 1: PADDY YIELD (IN KG)

Model	Split	RMSE	MAE	R ²	Adj. R ²
MLR	CV	950.5621	717.3783	0.9894	0.9893
	Test	948.7473	709.0288	0.9889	0.9887
Ridge	CV	950.5621	717.3785	0.9894	0.9893
	Test	948.7472	709.0289	0.9889	0.9887
SVR	CV	960.0052	707.6406	0.9892	0.9891
	Test	963.6298	705.5456	0.9886	0.9883
Random Forest	Train	832.8802	580.6327	0.9919	0.9918
	Test	850.4634	593.3191	0.9911	0.9909

TABLE V
MODEL PERFORMANCE — TARGET 2: PADDY YIELD PER HECTARE

Model	Split	RMSE	MAE	R ²	Adj. R ²
MLR	CV	224.4505	175.8662	0.3716	0.3630
	Test	223.3548	174.3518	0.3706	0.3423
Ridge	CV	224.3833	175.8629	0.3720	0.3634
	Test	223.4013	174.4068	0.3704	0.3420
SVR	CV	200.3499	147.5917	0.4996	0.4928
	Test	205.3706	150.2906	0.4679	0.4440
Random Forest	Train	176.9690	137.2620	0.6106	0.6064
	Test	206.0775	160.1625	0.4642	0.4401
XGBoost	Train	188.8607	146.9933	0.5565	0.5517
	Test	196.7047	152.9752	0.5119	0.4899
LightGBM	Train	188.7222	146.9379	0.5572	0.5524
	Test	196.6966	153.0358	0.5119	0.4899

near-equivalent results, with test R^2 values of approximately 0.37, confirming that the linear signal in the feature set is weak for this scenario. SVR algorithm with RBF kernel had a notable improvement by getting a CV R^2 of 0.4996 and a test R^2 of 0.4679. This validates the choice of a non-linear kernel for this target given the additional categorical and non-linear features. Among ensemble methods, XGBoost and LightGBM converged to identical test performance (R^2 being 0.5119, RMSE around 196.70). This demonstrates consistent generalization. Random forest achieved the highest training R^2 of 0.6106 but dropped to 0.4642 on the test set, indicating moderate overfitting in comparison to the gradient boosting models. The tighter train-test gap in performance of XGBoost and LightGBM suggests their regularization were better suited to the variance present in this target.

D. SHAP Analysis of Features Used in Best Models

For model explainability, SHAP (SHapley Additive exPlanations) was implemented on the best performing models for each target variable and their respective features. In this case, Random Forest Regressor for Target 1 and XGBoost for Target 2. Figure 1 demonstrates the results of the SHAP analysis. For random forest regressor model trained and tested on Target 1, the features 'LP-nurseryarea', 'DAP-20days', and 'Micronutrients-70Days' are the top three drivers. Higher

values of these features push predictions up, showcasing that they contribute to increased yield. This means that model has learned a good relationship between these inputs and target variable. Features towards the last (for example, 'Weed28D-thiobencarb') influence model predictions the least. .

For XGBoost model used for Target 2, the SHAP plot is noticeably much different. 'DAP-20days' is top of the feature importance ranks by a huge margin, with low values producing strongly negative SHAP contributions and high values showing positive contribution. The model heavily relies on early fertilizer application (DAP at 20 days) to predict yield. Categorical one-hot encoded features like 'Variety-ponmani' and 'Variety-delux ponni' are also on the higher side. The remaining features cluster near a SHAP value of zero, meaning the model draws little information from them during prediction. Fertilizer application at early growth stages and crop variety account for majority of the model's predictive output; all other variables contribute negligible. This concentration of predictive weight in a small feature subset may stem from correlations within the dataset or redundancy among the remaining variables. In summary, for Target 2, the model relies heavily on a couple agronomic input, whereas Target 1 benefits from a balanced feature contribution structure.

V. CONCLUSION

This study used a machine learning pipeline to predict rice yield from farm dataset of Tamil Nadu, India. two-stage feature selection process narrowed the original 45 parameters to specific subsets, and six regression models were evaluated across two target formulations, which are total yield and yield per hectare. Total yield prediction proved straightly, with all models exceeding R^2 of 0.98 and Random Forest performing best. The per-hectare target was considerably harder; XGBoost and LightGBM generalised most consistently at R^2 of 0.51, while Random Forest overfit. SHAP analysis confirmed that early agronomic inputs, particular DAP at 20 days, drive yield predictions most stronler, with weather feature contributing limited individual signal. The dual-target design yielded a clear finding: scale-normalised yield is harder regression problem, and model behaviour differs meaningfully between the two formulations. A limitation is the performance drop on Target 2, which points that the feature set does not fully capture per-hectare variance.

From a bigger perspective the abilty to forecast the yield of paddy crops by data-driven techniques may have the potential to facilitate better decision-making processes at both the farm level and the broader policy level. In areas like Tamil Nadu, where rice production is a livelihood factor for the population, the capacity to enhance the accuracy of yield predictions may be beneficial for farmers by helping them plan input usage and avoid risk more effectively. Although the results are still somewhat predicted the approach demonstrates potential of applying machine learning techniques to real-world data sets to identify key variables affecting crop yields



Fig. 1. SHAP summary plots for Total Paddy Yield using Random Forest (top) and Paddy Yield per Hectare using XGBoost (bottom).

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